

CHERGUELAINÉ Ayoub

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AI Engineer | GenAI | Data Engineer

AI Engineer specializing in **Generative AI, LLMs, NLP, and Data Engineering**, with strong software engineering expertise and hands-on experience building scalable AI systems. Experienced in **LLM fine-tuning, agentic AI, RAG, machine learning, document intelligence, and large-scale data processing pipelines**. Developed and fine-tuned open-source language models, while building production-ready AI and data solutions across backend, automation, and intelligent data processing. Passionate about turning advanced AI research into **practical, scalable, and production-grade systems**.

Work Experience

Paseetah - AI Engineer - 10/ 2024 - PRESENT

1. **Paseet, Riyadh** - 4 months - Part Time AI Engineer

- Implemented support for **geospatial interactions**, including adding and processing **polygon geometry** directly within the chat interface (property boundaries, zones, land selection, etc.)
- Fine-tuned open-source LLMs to build a **custom classification agent**, improving intent detection and property-related query categorization.

2. **Paseetah Platform – Data Quality, GeoAI, and Automation Project, Riyadh** – 12 months – AI & Data Engineer

- Improved the quality of legacy datasets (parcels, transactions, road) by cleaning, correcting missing information, and harmonizing inconsistent attributes.
- Designed and optimized **ETL/data pipelines** for large-scale real-estate and geospatial data, including scraping, cleaning, deduplication, validation, and performance optimization.
- Applied **Machine Learning, LLMs, OCR, and NLP** to real-estate and Arabic data, including transaction anomaly detection, Arabic text processing, translation, and information extraction.
- Developed **LLM agents and AI pipelines** to extract legal/regulatory rules from building codes and convert unstructured documents into structured, machine-readable data for automated compliance and downstream applications.

JURAI - Part-Time LLM Engineer & Full-Stack Developer - 1/ 2025 - 12/2025

1. **Jurai Assistant, Denmark** - 12 months - LLM Engineer & Full Stack Developer

- Designed and implemented **production-grade agentic AI systems** using **LangChain and CrewAI**, including retrieval, classification, reasoning, legal analysis, RAG, and decision-prediction agents.
- Built and integrated **LLM pipelines, semantic retrieval, embeddings, and long-term contextual memory**, with automated legal-data ingestion, preprocessing, validation, and enrichment pipelines.
- Developed the **FastAPI backend and AI orchestration layer**, integrating PostgreSQL/vector databases, external APIs, authentication, and scalable asynchronous workflows.

- **Deployed, tested, and continuously monitored AI services on AWS EC2 using Docker**, implementing production testing, logging, error handling, fallback mechanisms, performance monitoring, and iterative model/agent improvements.

BuySell - FrontEnd Developer - 2/ 2024- 12/2024

1. **BuySell Platform , Blida** - 12 months - FrontEnd Developer
 - **Developed and improved front-end features** for the Admin Platform using Next.js, TypeScript, and Tailwind, including dashboards, forms, tables, and management interfaces.
 - **Integrated and tested APIs** across admin workflows, ensuring reliable data flow and improving overall UI/UX performance.
 - **Built key e-commerce features** on the main platform such as product pages, category listings, search, filters, cart interactions, and navigation improvements.
 - **Optimized performance and code quality** through component refactoring, responsive design, and best practices to enhance speed, usability, and maintainability across both platforms.

SIG Services - Python Developer - 9/2023- 12/2023

1. **Odoo ERP Platform , Blida** - 3 months - Python Developer
 - Designing and implementing **Odoo modules** to meet business requirements, API Development & Testing

Icosnet - Artificial Intelligence Intern - 1/2023- 7/2023

2. **Document Classification Drive, Algiers** - 7 months - Artificial Intelligence Intern
 - Create an **intelligent system** utilizing transformers for accurate and **context-based classification** of company documents.
 - Developed an intelligent system base on context for **company documents classification**.

Education

- **Master degree on NATURAL LANGUAGE PROCESSING (Artificial Intelligence)** –
Saad Dahleb Blida University, Blida, Algeria - 09/ 2021- 06/ 2023
- **Bachelor degree on INFORMATION SYSTEMS AND SOFTWARE ENGINEERING** –
Saad Dahleb Blida University, Blida, Algeria - 09/ 2018 - 06/ 2021

Certificates

Building Transformer-Based Natural Language Processing Applications[[Details](#)]

Fundamentals of Deep Learning [[Details](#)]

Skills

Retrieval-Augmented Generation (RAG), AI Agents, LangChain , CrewAI, vLLM, n8n, Embeddings, Vector Stores, LLMs, Fine-tuning, Workflow Automation, Agent Orchestration, API Development, PostgreSQL, MongoDB, Redis, Data Scraping, Data Pipelines, Data Visualization, PyTorch, TensorFlow, Hugging Face, Transformers, Scikit-learn, Jupyter, FastAPI, Flask, [Next.js](#), React, Tailwind CSS, shadcn/ui, Docker, AWS EC2, CI/CD, Git, Google Cloud, Hugging Face, Python, TypeScript, JavaScript, SQL, Bash

Projects

- **Machine Learning & LLMs**

- Developed document-classification models using transformer architectures including [BERT](#), [BART](#), and [GPT-2](#) for real-world NLP tasks.
- Fine-tuned open-source LLMs including Qwen, LLaMA, and other architectures for domain-specific NLP, code generation, and instruction-following.
- Developed a collection of [Saudi Arabic dialect models](#) for dialect-specific NLP and conversational applications.
- Full fine-tuned [LFM2.5-2.6B Fable5 Coding Agent](#) on an agentic coding dataset, creating a lightweight coding-agent model optimized for local inference.
- Built lightweight LLM applications using [Gemini](#) and [Phi-3](#).
- Created and published a labeled [document-classification dataset](#) for supervised NLP training and evaluation.

- **Software Development**

- Developed [Noble Spending](#), a full-stack personal finance application built with **Next.js**.
- Built and deployed an open-source [OCR service](#) using **FastAPI** and open-source computer-vision/OCR models.
- Developed a self-hosted [Drive-like document management platform](#) with AI-powered document classification and a full-stack interface.
- Developed automated scraping and comparison systems for gaming marketplace deals, including data collection, processing, and price comparison.
- Developed a private web application for managing **importation and LC (Letter of Credit) workflows** in Algeria.
- Built additional private projects involving **REST APIs, web applications, automation, data processing, and AI integration**.